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Introduction to IP Addressing

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review



Learning Objectives

By the end of this lesson you will...

- Describe the basic concepts of **IP addressing** and its significance.
- Identify and explain the different **IP classes** and their uses.
- Explain the role of a **default gateway** in a network and its necessity.
- Describe **subnet masks** and their function in IP addressing.

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Key Vocabulary

- **IP Address.** A unique identifier assigned to each device on a network, allowing it to communicate.
- **Default Gateway.** A device that routes traffic from a local network to other networks, enabling external communication.
- **Subnet Mask.** A number that defines the network and host portions of an IP address, helping to organize networks.
- **IP Classes.** Categories (A, B, C, D, E) that classify IP addresses based on their size and intended use.

Previous Session review



Understanding Number Systems and ASCII



Learning Objectives

By the end of this lesson you will...

Convert between **decimal**, **binary**, and **hexadecimal** systems.

Explain the significance of **ASCII** in computing.

Describe how **hexadecimal** is used in technology.



Key Vocabulary

Decimal: A base 10 number system that uses digits from 0 to 9.

Binary: A base 2 number system that uses only the digits 0 and 1.

Hexadecimal: A base 16 number system that uses digits 0-9 and letters A-F.

ASCII: A character encoding standard for electronic communication.

Conversion: The process of changing a number from one system to another.



Poll

Which number system do you use most often?

- ☐ Decimal
- ☐ Binary
- ☐ Hexadecimal
- ☐ I use all of them!

Introduction to Number Systems

Today, we will explore different **number systems**: **decimal**, **binary**, and **hexadecimal**. Each system has its own unique way of representing numbers!



Decimal System

The **decimal system** is our everyday number system. It is a **base 10** system, which means it uses the digits **0 through 9**.



Binary System

The **binary system** is essential for computers. It operates on a **base 2** system, which means it only utilizes the digits **0** and **1**. This system allows computers to perform complex calculations and process data efficiently.



Hexadecimal System

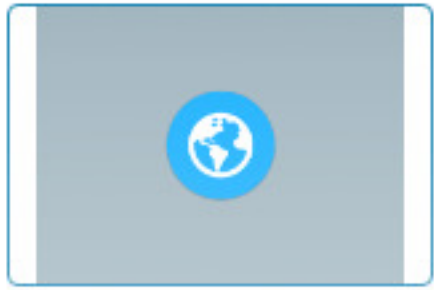
The **hexadecimal system** is a **base 16** system. It uses the digits **0-9** and the letters **A-F** to represent values.



Applications of Number Systems

Number systems are crucial in computing, data representation, and programming. They help us understand how computers process **information**.





Number Systems Discussion

Let's discuss how each number system is used in technology. Share your thoughts!

^ Instructions



Collaborate Board

Number Systems Discussion

Converting Decimal to Binary

To convert a decimal number to binary, we can use a step-by-step process. Here are the main steps involved:

Identify the decimal number you want to convert.

Divide the number by 2 and record the quotient and the remainder.

Continue dividing the quotient by 2 until it reaches 0, recording each remainder.

The binary number is formed by reading the remainders from bottom to top.



Converting Binary to Decimal

Understanding how to convert **binary** back to **decimal** is just as important. This process allows us to interpret **binary numbers** in a way that is meaningful and applicable in various contexts, such as computing and digital systems. We'll explore the reverse process!



Converting Decimal to Hexadecimal

Converting decimal to hexadecimal involves a few steps. This process allows us to change numbers from the decimal system, which is base 10, to the hexadecimal system, which is base 16. Here are the steps to follow:

Divide the decimal number by 16.

Record the remainder.

Continue dividing the quotient by 16 until it reaches zero.

Write the remainders in reverse order to get the hexadecimal value.



Converting Hexadecimal to Decimal

To convert **hexadecimal** numbers back to **decimal**, it is essential to understand the **place values**. Each digit in a hexadecimal number represents a power of 16, starting from the rightmost digit, which is the 16^0 place. This understanding helps in accurately converting the values.



Significance of Hexadecimal

The **hexadecimal** system is essential in various fields, particularly in programming and digital technology. It helps to represent complex data in a simpler format.

Used in **programming** to simplify code representation.
Essential for **memory addresses** in computer systems.
Commonly used in **color codes** for web design.



Multiple Choice

What is the decimal equivalent of the binary number 1010?

- ☐ 12
- ☐ 10
- ☐ 8
- ☐ 2

What is the decimal equivalent of the hexadecimal number 1A?

- ☐ 30
- ☐ 16
- ☐ 26
- ☐ 20

What is the binary representation of the decimal number 5?

- ☐ 111
- ☐ 110
- ☐ 100
- ☐ 101

Convert the binary number 1111 to decimal.

- ☐ 16
- ☐ 14
- ☐ 10
- ☐ 15

What is the decimal equivalent of the binary number 11001?

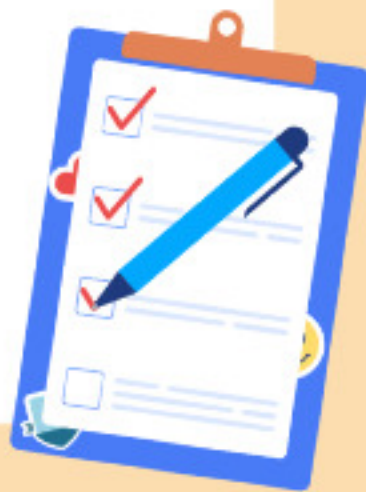
- ☐ 20
- ☐ 26
- ☐ 25
- ☐ 24

Convert the decimal number 30 to binary.

- ☐ 11100
- ☐ 11010
- ☐ 10110
- ☐ 11110

Hexadecimal in IT

In IT, **hexadecimal** makes it easier for humans to read and understand **binary data**. It's a bridge between the two!



Examples of Hexadecimal Use

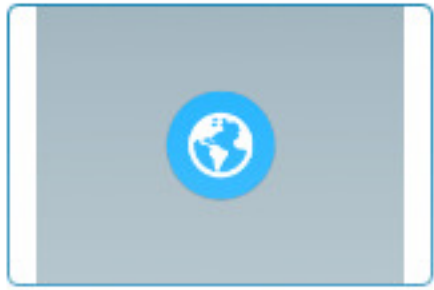
Hexadecimal is a base-16 number system that is widely used in various fields, particularly in web design and data representation. It plays a crucial role in creating visually appealing content.

Used in web design for **color codes**

Facilitates **data representation**

Helps create **visually appealing content**





Click on the Web link to investigate HTML Colour codes

HTML Colour Picker

^ Instructions



Collaborate Board

Click on the Web link to investigate HTML Colour codes

Open Ended Question

Explain why hexadecimal is important in computing. Think about its applications!

Please enter your answer here.

Multiple Choice

What is the hexadecimal representation of the decimal number 10?

- ☐ B
- ☐ D
- ☐ C
- ☐ A

Which of the following is a valid hexadecimal digit?

☐ G

☐ H

☐ F

☐ Z

What is the decimal equivalent of the hexadecimal number 1F?

- ☐ 32
- ☐ 31
- ☐ 30
- ☐ 29

How many digits are used in the hexadecimal system?

- ☐ 12
- ☐ 10
- ☐ 16
- ☐ 8

Which of the following pairs is equivalent in hexadecimal?

☐ 4D and 77

☐ 1A and 26

☐ 2B and 43

☐ 3C and 60

What is the hexadecimal number for the decimal number 255?

- ☐ F0
- ☐ FF
- ☐ FE
- ☐ F1

In hexadecimal, what does the letter 'C' represent in decimal?

- ☐ 12
- ☐ 10
- ☐ 11
- ☐ 14

What is the hexadecimal value of the decimal number 16?

- ☐ 1A
- ☐ 10
- ☐ 16
- ☐ 0F

What is the sum of the hexadecimal numbers A and 5?

- ☐ F
- ☐ C
- ☐ E
- ☐ D

Which of the following is not a hexadecimal number?

- ☐ 1A3
- ☐ G12
- ☐ 3E
- ☐ B4F

What is ASCII?

ASCII stands for **American Standard Code for Information Interchange**. It's a way to represent **text** in **computers**!



Purpose of ASCII

ASCII is a crucial standard that enables computers to communicate text effectively. It plays a vital role in data exchange and ensures compatibility between different systems.

ASCII stands for American Standard Code for Information Interchange. It represents text in computers and other devices that use text. ASCII ensures that different systems can understand the same text.



Importance of ASCII

ASCII is crucial for digital communication. It ensures that different devices can understand the same text!



Poll

How important do you think ASCII is in digital communication?

- ☐ Very important
- ☐ Somewhat important
- ☐ Not important

Multiple Choice

What does ASCII stand for?

- ☐ Advanced Standard Code for Information Interchange
- ☐ American Standard Code for Information Interchange
- ☐ American Standard Code for Internal Communication
- ☐ American Standard Code for Internet Integration

Which of the following characters is represented by the ASCII code 65?

- ☐ A
- ☐ B
- ☐ D
- ☐ C

What is the purpose of ASCII?

- ☐ To encode images
- ☐ To encode text characters
- ☐ To encode video
- ☐ To encode audio

What is the ASCII code for the space character?

- ☐ 31
- ☐ 33
- ☐ 30
- ☐ 32

Is the ASCII code for the letter 'A' greater than that for the letter 'B'?

☐ False

☐ True

True or False: ASCII can represent both letters and numbers.

☐ False

☐ True

Which of the following characters is NOT represented in ASCII?

- ☐ Uppercase letters
- ☐ Emoji characters
- ☐ Lowercase letters
- ☐ Digits 0-9

What is the purpose of ASCII?

- ☐ To store audio files
- ☐ To encode text for computers
- ☐ To encrypt data
- ☐ To create graphics

True or False: ASCII can only represent English characters.

- ☐ True
- ☐ False

What is the decimal value of the ASCII character 'A'?

- ☐ 32
- ☐ 90
- ☐ 97
- ☐ 65

Lesson Summary and Call to Action

In this lesson, we explored the process of converting numbers and discussed the significance of ASCII in our digital world. Here are the key points to remember:

Understanding how to convert numbers between different bases.

Recognizing the role of ASCII in representing characters digitally.

Identifying practical applications of number conversion in technology.

